Fiscal Year:	FY 2006	**Task Last Updated:**	FY 04/12/2006
PI Name:	Chaikin, Paul M. Ph.D.		
Project Title:	The Control and Dynamics of Hard Sphere Colloidal Dispersions		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:	NOTE: PI moved to NYU (from Princeton U) in 2005 per A. Hollingsworth in PI's dept (7/2009). Changed email 3/30/2009 (chaikin@princeton.edu no longer valid).		
Project Type:	Flight	**Solicitation / Funding Source:**	98-HEDS-03
Start Date:	01/14/2004	**End Date:**	01/13/2009
No. of Post Docs:	1	**No. of PhD Degrees:**	3
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Meyer, William V	**Contact Phone:**	216-433-5011
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Flight Program:			
Flight Assignment:	note: grant renewed 1/2004 for 4 years/new nag#		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Russel, William (Princeton)		
Grant/Contract No.:	NNC04GA60G		
Performance Goal No.:	1.23		
Performance Goal Text:	Pursue ground-breaking basic research in fluid physics and transport phenomena to provide fundamental understanding of natural phenomena affected by gravity for increased efficiency and effectiveness of space-based and industrial processes.		

Task Description:	This experiment consists of microscopy, microscopic and mesoscopic manipulation and control of colloidal samples to probe the essential features of the hard sphere disorder-order transition and the properties of the ordered solid phase that results. Many samples, in custom-designed microscope slide style cells that permit many different volume fractions and the application of external fields such as temperature gradients, periodic pattern imposition, and force application through the use of optical tweezers, will permit measurements from the disordered fluid phase, through the coexistence region, into the fully crystalline solid, and beyond into a glass-like phase. The dispersions will consist of colloidal hard spheres, in the 0.65 to 1.0 micron diameter range, with controlled polydispersity, consisting of poly (methyl methacrylate) (PMMA) in an index-matched fluid. Microscopy will allow for the observation of the position of individual colloidal particles and thus the determination of local crystalline or liquid structure, defect structure, observation of nucleation and growth of individual crystallites, the spontaneous formation of dendrites, the effect of periodic and other imposed patterns, field induced local melting and freezing, and the dynamics of liquid-crystal, liquid-glass and solid-glass interfaces. The applied fields will allow for controlled nucleation and growth, induced metastable phases, susceptibility, and rheology measurements. A simple design with many 50 microliter samples in a single sample holder on the microscope platen will allow interchange between the many samples to be studied. Static light scattering and confocal microscopy (replacing dynamic light scattering) will be done on the same samples. The static scattering image will be acquired with a CCD for Bragg scattering and structural studies. Confocal microscopy will allow the 3-d study of local affects. We expect to determine the volume fractions at which a fluid changes to a crystal, find the crystal structure of the equilibrium solid phase, contrast nucleation and growth in gravity and microgravity, especially in the region of the ground based glass transition (which we found does not occur in microgravity), quantitatively test the dendritic growth model, explore the role of polydispersity on the phase diagram and the structure of the different phases and explore the susceptibilities to external fields and the microrheology of the different phases. Characterization of crystal formation in the microgravity environment can lead to a greater understanding of how gravity affects many kinds of colloidal materials. Understanding these complex materials should enable new ways of forming ordered phases, such as those sought for photonic devices. The experiment employs a material that assumes multiple phases, which depend only on concentration. The simplest colloidal dispersion consists of hard spheres, which interact through hydrodynamic and Brownian forces, but feel no direct force before an infinite repulsion at contact. The hard sphere phase diagram, including the disorder-order transition, has been well established, but many-body interactions complicate the understanding of the kinetics of the transformations and the mechanical properties of disordered/metastable fluids and the ordered solid.Our experiments on these systems in microgravity, where the effects of sedimentation and convection are absent, have yielded several pleasant surprises. These included crystals (that were glasses on earth) and dendritic growth. In addition, Bragg images during nucleation and growth confirm some anticipated but previously unobserved features of the crystal growth process.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	

Task Progress: (11/01/04-12/31/05)
Two Ph.D. dissertations, either partially or fully supported by NASA funding, were completed during this period.

Ph.D. dissertation: “Photonic Quasicrystals and Random Ellipsoid Packings: Experimental Geometry in Condensed Matter Physics.” Weining Man

Abstract: This thesis investigates photonic quasicrystals and ellipsoid packings, two problems in which geometry plays an important role in condensed matter physics and where the use of stereolithographically-fabricated macroscopic models helps to solve fundamental problems.

Part I: Photonic quasicrystals may have bandgap properties that make them well-suited for scientific and technological applications in which photonic crystals are normally considered. They are constructed from two or more types of dielectric material arranged in a quasiperiodic pattern whose rotational symmetry is forbidden for periodic crystals. Because quasicrystals have higher point group symmetry (are closer to spherical) than ordinary crystals, their gap center frequencies are closer and the gap widths are more uniform. These are optimal conditions for forming a complete, more spherical bandgap. Our calculations on 2-D Penrose quasicrystals show very isotropic bandgaps. Analogous computations on 3-D quasicrystals are challenging and have not been performed to date. Here we circumvent the 3-D computational problem by doing an experiment. Using stereolithography, we constructed an icosahedral photonic quasicrystal with centimeter-scale cells and performed microwave transmission measurements. We show that the quasicrystal exhibits sizeable stopgaps and determine their effective Brillouin zones. As a check, we perform calculations and experiments on a similar photonic crystal with diamond symmetry. Our studies confirm that quasicrystals are excellent candidates for photonic bandgap materials.

Part II: Recent simulations indicate that ellipsoids can randomly pack more densely than spheres and, remarkably, ellipsoids with an aspect ratio of 1.25 : 1 : 0.8 can approach the densest crystal packing (FCC) of spheres, with a packing fraction of 74%. We demonstrate that such dense packings are experimentally realizable. We introduce a novel way of determining packing density for a finite number of particles that minimizes wall effects. We have fabricated ellipsoids using stereolithography and show that in a spherical container of radius R, the radial packing fraction, f(r), can be obtained from V(h), the volume of added fluid to fill the sphere to height h. Additionally, we perform an MRI scan of our packing and obtain f(r) from it. The measurements of the overall density and the core density were 0.74 ± 0.005. This value agrees with our simulations.

Ph.D. dissertation: “Confocal microscopy of hard sphere colloidal suspensions.” Matthew T. Sullivan

Abstract: This thesis investigates the formation and rheology of hard sphere crystals using confocal microscopy. Our experiments employ colloidal dispersions of poly(methyl methacrylate) (PMMA) particles stabilized by poly(hydroxystearic acid) (PHSA) that exhibit hard sphere behavior. These systems have been studied extensively using light scattering and bulk rheology; however, microscopy offers many important experimental advantages over these conventional techniques. Here we introduce a method for controlling density in soft systems using applied external fields, useful for small volumes. A quantitative theory is developed to exactly predict concentrations in a given system. Measurements on density-matched hard sphere colloidal particles confirm this theory and the technique allows us to produce large colloidal crystals in a controlled manner. Then, we use confocal microscopy to follow the three dimensional growth from a patterned template surface designed to nucleate and grow oriented crystals. The growth can be understood theoretically, allowing us to measure the kinetic prefactor in colloidal crystallization. We observe that the

Task Progress:

crystallization front is rough on the scale of the colloidal particles. The growth rate as a function of the calculated mean curvature can be used to measure the fluid-solid surface tension. Lastly, the elastic response of the resulting crystal can be measured from cross-correlations of thermally driven particle fluctuations. From the radial and angular dependence of the response, the full elastic tensor can be measured.

Highlights from chapter 2: An electric bottle for colloids.

The behavior of colloidal and soft systems is influenced by a number of parameters such as temperature, pressure, and density. One of the most important for determining phase behavior is the concentration, or alternatively the volume fraction. Unlike temperature, there is no direct way of varying volume fraction, so changing it generally requires producing individual samples at each desired volume fraction. In this chapter, we discuss a novel method for directly controlling particle volume fraction using “dielectrophoretic equilibrium”. We use a finite width parallel-plate capacitor to create an “electric bottle”, with particles uniformly trapped in the center of the bottle at higher volume fraction than the surrounding suspension. Using a model system of colloidal hard spheres, we demonstrate the effectiveness of this technique and can observe liquid, crystal, and glassy phases. We develop a simple procedure for calculating equilibrium volume fraction and quantitatively verify our predictions. We can create large density gradients as well as uniform regions, and can control the resulting density by tuning the applied potential. The mechanism and thermodynamics of the system are well understood and can be accurately predicted using only dielectric properties, the electric field behavior, and the system equation of state. While induced interparticle interaction effects can be important, we demonstrate they can be negligible even for large densification and discuss a technique for identifying when they become important. Highlights from chapter 3: Microrheology of hard sphere FCC crystals.

The mechanical properties of soft materials are often anisotropic, but measurements using conventional techniques are not suited to measuring that anisotropy. We extend the methods of conventional microrheology to deal with anisotropic materials. In particular, we use microrheology to measure the full elastic tensor of an FCC hard sphere crystal. Using a surface template and density matched hard sphere-like suspensions, we have measured the three elastic constants of this particular type of crystal. Our measured values agree reasonably well with values from density functional theory calculations and simulation.

Highlights from chapter 4: Epitaxial Growth Kinetics.

Three dimensional confocal microscopy was used to measure the growth rates of colloidal hard sphere crystals from templated surfaces that nucleate growth along the FCC (100) plane. We first discuss classical nucleation and growth theory and adapt them to growth from a surface template. Then, we present the results of experiments performed on hard sphere colloidal crystals to measure the growth rate on unstrained lattices. Using this technique, we measure both the kinetic prefactor in the growth of colloidal hard sphere crystal as well as the fluid-solid surface tension. After an initial onset, the growth is observed to be steady throughout the experiment and can be described in terms of the classical Wilson-Frenkel growth law. Using the interfacial volume fraction and known values for the self-diffusion and chemical potential, we can measure the numerical prefactor to this expression for colloidal hard spheres. In measuring the sample volume fraction from the interface, we detected a region of depleted volume fraction in front of the crystal interface. While this has long been suggested to exist in colloidal systems, this is the first known measurement of the depleted volume fraction region in a growing colloidal crystal.

Properties and manipulation of colloidal platelets in electric fields (K. Zhao).

We have investigated the phase behavior of hard rectangles consisting of photolithographically prepared, plate-like colloidal PMMA disks (diameter = 5.3 microns, thickness = 0.8 microns) standing on edge. The disk monolayer adopts this particular orientation due to an applied electric field normal to the cover slip. Our model system thus resembles a two dimensional set of colloidal rectangles. We have detected a Kosterlitz–Thouless phase transition from isotropic to nematic (quasi-smectic). Between these two phases, we find a regime where tetratic correlations are longer range than nematic. By studying the disclinations and domain walls, we suggest that the tetratic phase is driven by the nearby nematic and exists on a length scale larger than the nematic domain wall spacing but smaller than the interdisclination distance.

PHSA-g-PMMA comb graft stabilizer synthesis (A.D. Hollingsworth). Sterically-stabilized colloidal particles are an excellent model hard-sphere system used by many groups. One of the original stabilizers used for such systems was developed and patented by ICI (UK) more than 30 years ago. It consists of a ‘comb-like’ stabilizer of a poly(12-hydroxystearic acid) (PHSA) which is soluble in aliphatic hydrocarbons. These pendant PHSA chains are grafted to an insoluble poly(methyl methacrylate) backbone that strongly adsorbs to polymer particles and thus provides a means of anchoring stabilizer to particle surfaces. Unfortunately, the PHSA-g-PMMA stabilizer is not commercially available. Furthermore, the three-step procedure (Antl, et al. 1986) is generally regarded as technique intensive, time-consuming, and somewhat vague. We have systematically studied the stabilizer synthesis with the goal of making the entire protocol understandable as well as reproducible. Several important details, not published in the literature, were determined using mass spectroscopy, proton NMR, acid titration and gel permeation chromatography (GPC). The analytical results were used to characterize the polymer and its precursors.

The starting material for the soluble PHSA chains is 12-hydroxystearic acid (12-HSA). It is generally contaminated with stearic acid which is difficult to separate. Stearic acid participates in the polymerization but produces dead chains, which limits molecular weight. Four to five ester linkages are required to produce a ~10 nm steric layer on the particle surfaces. Water is a byproduct of the polycondensation reaction and must be removed for any reasonable conversion to PHSA. Water removal from the reactor is accomplished via heteroazeotropic distillation using a special glass receiver. A technical grade (nominally 70%) 12-HSA was polymerized to the target number average molecular weight of 1700, whereas a 90+% grade hydroxy acid produced a polymer with three times this molecular weight. A commercial PHSA (uniqema Hypermer LP1) was also tested and found to produce satisfactory results.

In the second step of the stabilizer synthesis, a difunctional connector molecule, glycidyl methacrylate, was reacted with the terminal carboxyl groups on the PHSA chains via a ring opening, SN2 reaction. The unsaturated part of the monomer was protected with an inhibitor. Lastly, we copolymerized the macromonomer from step two with methyl methacrylate (anchor monomer) and additional glycidyl methacrylate. The reactor was starve fed to ensure product uniformity, i.e., random copolymer not homopolymer, and to suppress the anchor chain molecular weight. Using the stabilizer synthesized from the uniqema PHSA, we were able to reproducibly prepare low polydispersity PMMA particles. We believe that the PHSA-g-PMMA stabilizer supply problem is under control now and plan to publish a paper detailing the results.

	University of Michigan particle synthesis workshop Aug. 15-17, 2005: Mike Solomon's group (Chemical Engineering) organized an informal workshop to share their four-year experience with PMMA particle synthesis. A.D. Hollingsworth attended the 2-1/2 day program along with Carlos van Kats (van Blaaderen group), Mark Buckley (Itai Cohen group, Cornell U.), Alex Meyer (Furst group, U. of Del.), Gianguido Cianci (Weeks group, Emory U.), as well as three graduate students from the Solomon group. During the workshop, the Michigan group attempted to make the stabilizer, as well as PMMA particles. Solomon's students started from scratch, using a technical grade 12-HSA, but the Stage III polymerization failed because one of the monomers polymerized in the addition funnel. However, we did produce micron-sized, monodisperse particles from another batch of 'homemade' stabilizer. The workshop was well organized and a handbook detailing the entire particle and stabilizer synthesis was distributed.
Bibliography Type:	Description: (Last Updated: 09/29/2025)
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